

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Industry Problem solving task
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	BE ECE Final year	Date:	17/08/2026

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage in handling these workloads. Propose a memory hierarchy that reduces access latency, improves data locality, and supports high concurrent transaction throughput.
2. A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate the suitability of SRAM, DRAM, and non-volatile memory for storing frequently accessed control data, sensor information, and program code. Design a memory hierarchy that minimizes latency and ensures predictable memory access for real-time operations.
3. A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze how cache size, DRAM capacity, memory bandwidth, and storage performance influence video-processing workloads. Recommend a hierarchical memory organization that improves data transfer rates, reduces processing delays, and supports smooth real-time editing.
4. A weather forecasting system performs complex numerical calculations on massive datasets containing atmospheric temperature, pressure, wind, and humidity measurements. Analyze how cache hierarchy, DRAM bandwidth, and large-capacity memory influence the performance of repeated numerical computations. Propose memory-hierarchy enhancements that improve data locality, reduce memory access bottlenecks, and increase computational throughput.
5. A smart home gateway manages data from cameras, motion sensors, environmental sensors, and connected appliances while operating continuously with limited energy resources. Compare SRAM, DRAM, Flash, and other suitable memory technologies based on latency, capacity, persistence, and power consumption. Design an efficient memory hierarchy that enables rapid access to frequently used data while minimizing energy consumption and maintaining reliable long-term data storage.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism(5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 25	/ 15	/ 10	/5	/ 100

Course Coordinator**Faculty Incharge**

Q1. Memory Hierarchy for E-Commerce Platform

1. Problem Understanding

During peak shopping periods, thousands of users access product catalogs, customer records, and transaction data simultaneously. This increases memory traffic and response time. The system therefore needs low latency, high bandwidth, and high transaction throughput.

2. Role of Memory

- **Cache (L1, L2, L3):** Stores frequently accessed data and instructions, reducing CPU access time.
- **DRAM:** Stores active product catalogs, customer sessions, transaction data, and running applications.
- **Secondary Storage (SSD/NAND Flash):** Stores large, persistent data such as databases, transaction history, logs, and backups.

3. Proposed Memory Hierarchy

CPU

|

L1 Cache → L2 Cache → L3 Cache

|

DRAM

|

SSD / NAND Flash

4. Optimization

- Cache frequently accessed product information.
- Use database indexing to improve data locality.
- Keep active database pages in DRAM.
- Use high-speed SSD for persistent storage.
- Reduce cache misses and unnecessary data transfers.

5. Conclusion

The **L1–L2–L3 Cache → DRAM → SSD** hierarchy reduces access latency, improves data locality, and supports high concurrent transaction throughput by placing frequently used data in faster memory. This approach address the memory-hierarchy and performance requirements specified in the assessment.

Q2. Memory Hierarchy for a Robotic Manufacturing System

1. Problem Understanding

A robotic manufacturing system processes camera images, sensor readings, and control instructions under strict real-time deadlines. Therefore, it requires low and predictable memory latency.

2. Memory Technology Selection

- **SRAM:** Best for frequently accessed control data because it has very low latency.
- **DRAM:** Suitable for large camera images and sensor data because it provides higher capacity.
- **Non-volatile memory (Flash/MRAM):** Suitable for program code, firmware, and configuration data because it retains data without power.

3. Proposed Memory Hierarchy

CPU

|

SRAM / L1 Cache → Critical Control Data

|

L2/L3 Cache → Frequently Used Sensor Data

|

DRAM → Camera Images & Large Sensor Data

|

Flash / MRAM → Program Code & Firmware

4. Optimization

- Keep critical control data in SRAM/cache.
- Store large camera frames in DRAM.
- Keep firmware and configuration in non-volatile memory.
- Use data locality to reduce unnecessary memory accesses.
- Avoid unnecessary paging for real-time data.

5. Performance Analysis

Memory performance can be evaluated using:

- Latency=Access Time
- Bandwidth=Data Transferred/time

Lower latency provides faster and more predictable robotic control.

6. Conclusion

The proposed SRAM/Cache → DRAM → Flash/MRAM hierarchy provides low latency for critical control operations, sufficient capacity for sensor and camera data, and reliable storage for program code. This helps the robotic system meet its strict real-time deadlines.

Q3. Memory Hierarchy for a 4K/8K Video-Editing Workstation

1. Problem Understanding

A video-editing workstation processes multiple 4K and 8K video streams, which requires high memory bandwidth, sufficient DRAM capacity, large cache, and fast storage to reduce processing delays.

2. Role of Memory

- **Cache:** Stores frequently used video-processing instructions and data, reducing CPU access time.
- **DRAM:** Stores video frames, editing buffers, and intermediate processing data.
- **SSD/NAND Flash:** Stores large video files, project files, and rendered data with high-speed data transfer.

3. Proposed Memory Hierarchy

CPU / GPU

|

L1 → L2 → L3 Cache

|

High-Bandwidth DRAM

|

NVMe SSD / NAND Flash

4. Optimization

- Use larger cache for frequently reused data.
- Use high-bandwidth DRAM for video frames.
- Use high-speed SSD for source and project files.
- Use sequential data access to improve data transfer.
- Use prefetching to reduce waiting time.

5. Performance Analysis

Higher memory bandwidth allows more video data to be processed per second and reduces processing delays.

6. Conclusion

The **L1–L2–L3 Cache → High-Bandwidth DRAM → NVMe SSD** hierarchy provides fast data access, high transfer rates, and sufficient storage capacity, making it suitable for smooth **4K/8K real-time video editing**.

Q4. Memory Hierarchy for a Weather Forecasting System

1. Problem Understanding

A weather forecasting system processes massive datasets containing **temperature, pressure, wind, and humidity**. Repeated numerical calculations require high memory bandwidth, good data locality, and large memory capacity.

2. Role of Memory

- **L1/L2 Cache:** Stores frequently reused numerical data and instructions, reducing access latency.
- **L3 Cache:** Stores shared data used by multiple processor cores.

- **DRAM:** Stores large weather datasets and provides high memory bandwidth.
- **Large-capacity storage:** Stores historical weather data and large datasets.

3. Proposed Memory Hierarchy

CPU

|

L1 Cache → L2 Cache → L3 Cache

|

High-Bandwidth DRAM

|

Large-Capacity Storage

4. Optimization

- Use temporal locality by reusing recently accessed data.
- Use spatial locality by storing related data close together.
- Use cache blocking/tiling for numerical calculations.
- Use high-bandwidth DRAM.
- Reduce unnecessary data transfers.

5. Conclusion

The proposed **L1–L2–L3 Cache → High-Bandwidth DRAM → Large-Capacity Storage** hierarchy improves data locality, reduces memory-access bottlenecks, and increases the computational throughput of weather forecasting calculations.

Q5. Memory Hierarchy for a Smart Home Gateway

1. Problem Understanding

A smart home gateway continuously manages camera data, motion sensors, environmental sensors, and connected appliances while operating with limited energy. Therefore, it requires low latency, low power consumption, sufficient capacity, and reliable storage.

2. Memory Technology Comparison

- **SRAM:** Very fast and suitable for frequently accessed or critical data.
- **DRAM:** Higher capacity and suitable for active sensor and camera data.
- **Flash:** Non-volatile and suitable for long-term data storage.
- **MRAM/RRAM:** Suitable for persistent data requiring faster access.

3. Proposed Memory Hierarchy

CPU

|

SRAM / Cache

|

DRAM

|

MRAM / RRAM

|

NAND Flash

4. Optimization

- Keep frequently accessed data in **SRAM/cache**.
- Store active camera and sensor data in **DRAM**.
- Use **MRAM/RRAM** for important persistent data.
- Use **NAND Flash** for long-term storage.
- Move inactive data to lower levels to reduce power consumption.
- Reduce unnecessary memory transfers.

5. Conclusion

The **SRAM/Cache → DRAM → MRAM/RRAM → NAND Flash** hierarchy provides a balance between **speed, capacity, persistence, reliability, and power consumption**. It enables rapid access to important smart-home data while providing efficient long-term storage.
